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Governments of the Republic of Korea, Japan, the People's Republic of China, the United Kingdom, the United States, France and Germany, with their legislatures, and the public development banks Bpifrance and KfW

Summary: This report reviews the entrepreneurial ecosystem policy that the governments of Korea, Japan, China, the United Kingdom, the United States, France and Germany set out from January to December 2022 in six domains — regulation and law, finance, culture, support, human capital and markets — and analyses it in its context, against theory and in practice. Finance was the centre of policy and culture the thinnest domain. The best measures lowered the cost of founding and of failure, as China's registration rules and France's sole-trader statute did, or put public money behind private managers, as Germany's Future Fund modules did; the weakest were plans, proclamations and awards, an announced cut to research relief for small firms and reforms left as bills or key points. Korea and France formed new governments, the United Kingdom changed its Prime Minister twice, and most of Japan's five-year plan was still to be enacted.

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Entrepreneurial Ecosystem Policy in Korea, Japan, China, the United Kingdom, the United States, France and Germany, 2022

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Entrepreneurial Ecosystem Policy in Korea, Japan, China, the United Kingdom, the United States, France and Germany, 2022

Abstract

This report reviews and analyses the entrepreneurial ecosystem policy of the governments of the Republic of Korea, Japan, the People's Republic of China, the United Kingdom, the United States, France and Germany in their plans, laws, budgets and announcements. The policies are reviewed in six domains — regulation and law, finance, culture, support, human capital and markets — and analysed in their context, against theories of entrepreneurial ecosystems, entrepreneurial finance, business failure and policy design, and in practice. Finance was the centre of policy. Where public money followed private choices, as in Korea's Fund of Funds and TIPS and Germany's Future Fund modules, the designs followed the evidence; the US reform of its investment companies remained a proposal, and France's First Factory call chose projects itself. China's registration rules, France's sole-trader statute and Germany's online formation lowered the cost of founding or of failure, while Japan's security right over a whole business remained a study and the British register reform awaited a bill. Personal guarantees divided policy: Japan announced a credit guarantee without one, while the British Recovery Loan Scheme let lenders require one. Talent policy rested on the British routes for graduates and the skilled staff of growing firms; Korea's and Japan's visa changes were plans, and Germany's immigration key points made no provision for founders. Culture was the thinnest domain, addressed by events, awards and proclamations. Korea and France formed new governments, the United Kingdom changed its Prime Minister twice and reversed most of its Growth Plan, and many reforms were plans, tax outlines, key points or bills at the end of the year; performance was reported as outputs, not effects. The period covered is January to December 2022.

Keywords

start-up definition, company registration, funds of funds, credit guarantees, personal guarantees, investor tax relief, stock options, start-up visas, restart support, public procurement, listing reform

JEL Classification

L26, L53, G24, H25, O38, M13

Entrepreneurial Support Organisation Research Division, Institute for Entrepreneurial Ecosystem Development (IFEED).

1 INTRODUCTION

Key findings

- Finance was the centre of policy, and culture remained the thinnest domain: governments held events, gave awards, proclaimed a week and a month and stated aims, and few measures touched attitudes to failure.
- The cost of founding and of failure fell where rules were in force. China's registration rules allowed suspension and simplified deregistration, France separated sole traders' business assets from personal ones and opened a one-stop shop, Germany allowed online formation from 1 August and Korea widened its definition of a start-up from 29 June; Japan's security right over a whole business remained a study, the British register reform awaited a bill, and US ownership reporting will add a duty for about 32.6 million companies.
- Public money worked best where private managers chose the firms. Korea's Fund of Funds offered KRW 929.7 billion in one call and its adopted budget raised TIPS to KRW 378.2 billion, and Germany launched EUR 1.2 billion of growth loans alongside private lenders; the US reform of its investment companies remained a proposal, and France's First Factory call chose 18 projects itself.
- Personal guarantees divided policy. Japan announced a credit guarantee without a personal guarantee, to start in March 2023, and France extended an allowance to independent workers whose business was not viable, while the British Recovery Loan Scheme let lenders require personal guarantees.
- Talent and investor reliefs were mostly proposals. The United Kingdom opened routes for graduates and the skilled staff of growing firms, but Korea's and Japan's visa changes were plans, Germany's immigration key points made no provision for founders, and Korea's higher stock-option ceiling, Japan's 15-year exercise period and the British seed-scheme limits were proposed or announced.
- Politics and money shaped the year. Korea's new government turned to private-led venture investment; the United Kingdom changed its Prime Minister twice, and its Autumn Statement reversed nearly all of the Growth Plan and will cut research relief for small firms; France formed a new government; Germany's budget for 2023 met its borrowing limit; and the US appropriations for the SBA were enacted on 29 December.
- Performance was reported as outputs, not effects, and the governments' own figures cut against their narratives: Korean venture investment in the third quarter was KRW 838.8 billion below a year earlier, British private businesses fell by 1.5 %, the number of small US federal vendors fell by almost 5.8 % as contract dollars reached a record, and only 12 % of innovative German start-ups sought venture capital.

The entrepreneurial ecosystem approach holds that productive entrepreneurship depends on many conditions at once — policy, finance, culture, supports, human capital and markets — and

that governments go wrong when they act on only one or two of them or try to emulate Silicon Valley (Isenberg, 2010, 2011). The approach has since been developed into models in which the conditions are interdependent and in which the weakest condition limits the performance of the system (Ács, Autio, & Szerb, 2014; Stam, 2015), and in which the relationships among cultural, social and material attributes reproduce the ecosystem (Spigel, 2017). This report applies it to Korea, Japan, China, the United Kingdom, the United States, France and Germany in one year. It asks three questions:

1. What policies did each of these governments set out in each domain of the entrepreneurial ecosystem in 2022?
2. How did the policies arise and how were they justified, how far were they valid in theory, and what had they achieved in practice by the end of the year?
3. What were their limitations, and what follows for policy?

The report draws on the policy studies and entrepreneurship literature where it bears on a policy — on problem definition and representation (Bacchi, 2009; Rochefort & Cobb, 1993), entrepreneurial ecosystems (Brown & Mason, 2017; Stam, 2015), government venture capital and loan guarantees (Brander, Du, & Hellmann, 2015; Lerner, 2009; Riding, Madill, & Haines, 2007), investor taxation (Edwards & Todtenhaupt, 2020), entry regulation (Klapper, Laeven, & Rajan, 2006), business failure (Armour & Cumming, 2008; Lee, Peng, & Barney, 2007; Lee, Yamakawa, Peng, & Barney, 2011), immigrant talent (Hunt, 2011; Kerr & Kerr, 2020) and public procurement (Edler & Georghiou, 2007).

Section 2 reviews the key policies of 2022 in six domains and assesses each country's policies in each domain. Section 3 analyses the policies in their context, against theory and in practice. Section 4 sets out the debates and contested points they raised and their theoretical limitations, and Section 5 their implications for policy.

The report rests on the plans, laws, budgets, reports and announcements in which these governments set out their entrepreneurial ecosystem policy in 2022: cabinet plans and economic measures, acts of the legislature and bills, budgets and budget requests, rules and decrees in official journals, statistics, and the press releases of ministries, agencies and the public development banks Bpifrance and KfW, acting for the state.

Table 1 lists the sources by country. Quotations from sources that are not in English are IFEED's translations.

Table 1: Government plans, laws, reports and announcements reviewed

Subject	Issuing bodies	Date and type	Period covered	Description
Korea				
Stock-option valuation decree	MSS	15 February 2022 Press release	February 2022	More ways to value unlisted shares; tax-free ceiling.
Fund of Funds second call	MSS	22 February 2022 Press release	February 2022	KRW 929.7 billion offered to form KRW 1.6 trillion of funds.
Restarter integrity evaluation	MSS	15 March 2022 Press release	2017–March 2022	One evaluating body; items cut from five to three.
Start-up Support Act	MSS	21 June 2022 Press release	June 2022	Act in force 29 June; start-up scope widened.
Ministry work plan	MSS	12 July 2022 Press release	July 2022	Multiple-voting shares bill pending; private-led turn.
Tax revision proposal	MOEF	21 July 2022 Proposal	July 2022	Stock-option ceiling; investment tax breaks extended.
Venture Investment Act decree	MSS	16 August 2022 Press release	August 2022	Accelerator funds; M&A holding; insiders' liability.
Ministry budget proposal for 2023	MSS	31 August 2022 Press release	August 2022	KRW 13.6 trillion; TIPS and Fund of Funds lines.
K-Startup global strategy	MSS	8 September 2022 Strategy	September 2022	Global partners, visas, overseas centres, brand.
Venture investment ecosystem plan	MSS	4 November 2022 Plan	January–November 2022	Third-quarter fall; private funds of funds.
Re-challenge Day	MSS	22 November 2022 Press release	November 2022	Event to spread a re-challenge culture.
Ministry budget for 2023	MSS	26 December 2022 Press release	December 2022	KRW 13.5205 trillion passed; TIPS to 700 teams.
Work plan for 2023; Venture Business Act	MSS	27 December 2022 Press releases	2019–December 2022	Unicorn rank; stock-option rules for experts.
Innovation-product trials	Public Procurement Service	28 December 2022 Press release	December 2022	58 products worth KRW 15.6 billion from 2023.
Japan				
Comprehensive economic measures	Government of Japan	28 October 2022 Cabinet decision	October 2022	First year of start-ups; fiscal spending.
Security right working group	FSA	2 November 2022 Meeting paper	November 2022	Security over a whole business proposed.
Start-up Development Five-year Plan	Council for New Capitalism	28 November 2022 Plan	November 2022	Investment target, funds, guarantee, visa, procurement.
Tax reform outline	Government of Japan	23 December 2022 Cabinet decision	December 2022	Reinvestment relief; stock options; open innovation.

Subject	Issuing bodies	Date and type	Period covered	Description
Management guarantee reform	METI; FSA; Ministry of Finance	23 December 2022 Programme	December 2022	Guarantee without a personal guarantee from 2023.
FILP plan for fiscal year 2023	Ministry of Finance	23 December 2022 Budget material	December 2022	Start-up loan ceiling doubled; DBJ fund.
Foreign investment interim summary	Cabinet Office	23 December 2022 Council paper	December 2022	Research start-up fund; start-up visa.
China				
Entrepreneurship driving employment	NDRC and others	8 February 2022 Notice	February–December 2022	Demonstration bases for graduates.
Venture and angel tax conditions	MOF; STA	9 February 2022 Announcement	January–December 2022	Size limits kept to the end of 2023.
Market-entity registration rules	SAMR	1 March 2022 Rules	March–December 2022	Electronic licences; suspension; simple exit.
Government work report	State Council	5 March 2022 Work report	2021–2022	Tax refunds, deficit, registration system.
Bank lending to small firms	CBIRC	6 April 2022 Notice	April–December 2022	Lending targets; pledges in place of real estate.
National incubators	MOST	19 April 2022 Notice	April 2022	149 incubators recognised.
Graduate employment and entrepreneurship	State Council General Office	5 May 2022 Notice	May–December 2022	Hiring and start-up subsidies; free space.
Hand-in-Hand Action	MIIT and others	12 May 2022 Notice	May–December 2022	Matching of large firms and SMEs; payment terms.
Package to stabilise the economy	State Council	24 May 2022 Notice	May–December 2022	Refunds, re-guarantees, procurement, SRDI firms.
Procurement support for SMEs	MOF	30 May 2022 Notice	July–December 2022	Price preference and reserved shares raised.
Gradient cultivation of SMEs	MIIT	1 June 2022 Measures	June–December 2022	Three tiers of quality SMEs; targets.
Decree No. 755	State Council	1 October 2022 Regulation	November–December 2022	Individual businesses; in force 1 November.
United Kingdom				
Finance Act 2022	Parliament	24 February 2022 Act	February–December 2022	GBP 1 million allowance to March 2023.
Register reform white paper	BEIS	28 February 2022 White paper	February 2022	Identity checks; registrar's powers.
Immigration rules HC 1118	Home Office	15 March 2022 Memorandum	March–December 2022	High Potential Individual and Scale-up routes.
Queen's Awards for Enterprise	BEIS	21 April 2022 Press release	April 2022	Awards to 232 businesses.
Procurement Bill	Cabinet Office	12 May 2022 News story	May 2022	Duty to consider SMEs; 30-day payment.

Subject	Issuing bodies	Date and type	Period covered	Description
Help to Grow: Management	BEIS; HM Treasury	14 June 2022 Press release	June 2022	Two participants for larger small firms.
Recovery Loan Scheme	BEIS	20 July 2022 Press release	July 2022	Two more years; personal guarantees allowed.
Mansion House speech	HM Treasury	20 July 2022 Speech	July 2022	Risk takers; capital-market reforms.
Growth Plan 2022	HM Treasury	23 September 2022 Fiscal statement	September 2022	Seed scheme limits; red tape; Investment Zones.
Start Up Loans widened	BEIS	25 September 2022 Press release	June 2012–September 2022	Firms up to three years; second loans.
Business population estimates	BEIS	6 October 2022 Statistics	2021–2022	5.5 million businesses, 1.5 % fewer.
Ministerial appointments	Prime Minister's Office	14 October 2022 Announcement	October 2022	Jeremy Hunt appointed Chancellor.
Ministerial appointments	Prime Minister's Office	25 October 2022 Announcement	October–November 2022	Rishi Sunak Prime Minister; Hunt re-appointed.
Autumn Statement 2022	HM Treasury	17 November 2022 Fiscal statement	October–November 2022	Growth Plan reversed; R&D relief for SMEs cut.
Edinburgh Reforms	HM Treasury	9 December 2022 Release, policy note	December 2022	Prospectus threshold dropped; offer platforms.
United States				
Community Advantage pilot	SBA	29 April 2022 Notice	May–December 2022	Changes in force from 31 May.
National Small Business Week	President	29 April 2022 Proclamation	May 2022	Almost 5.4 million applications stated.
Loan programme burden rule	SBA	30 June 2022 Final rule	August–December 2022	Obsolete loan rules removed from 1 August.
Prime contracting fact sheet	SBA	July 2022 Fact sheet	October 2020–September 2021	USD 154.2 billion to small firms; fewer vendors.
CHIPS and Science Act	US Congress	9 August 2022 Act	August–December 2022	Fellowships; entrepreneurial leave; lab programmes.
Inflation Reduction Act	US Congress	16 August 2022 Act	August–December 2022	Payroll offset of the research credit raised.
SBIR and STTR Extension Act	US Congress	30 September 2022 Act	September–December 2022	Programmes extended; foreign-risk checks.
Beneficial ownership reporting	FinCEN	30 September 2022 Final rule	September–December 2022	Owner reporting for 32.6 million companies.
SBIC proposed rule	SBA	19 October 2022 Proposed rule	October–December 2022	Accrual Debenture; licensing fees.
Affiliation and lending proposal	SBA	26 October 2022 Proposed rule	October–December 2022	Affiliation by ownership; insurance threshold.

Subject	Issuing bodies	Date and type	Period covered	Description
National Entrepreneurship Month	President	31 October 2022 Proclamation	November 2022	A record 5.4 million new businesses stated.
SBLC moratorium proposal	SBA	7 November 2022 Proposed rule	November–December 2022	New lending companies; mission-based type.
SBDC proposed rule	SBA	13 December 2022 Proposed rule	December 2022	First revision since 1995; client opt-in.
Consolidated Appropriations Act, 2023	US Congress	29 December 2022 Act	December 2022	SBA programmes and loan ceilings enacted.
SBA budget request	SBA	2022 Budget request	2021–2022	Requests for development programmes.
France				
Independent activity Act	Parliament	14 February 2022 Act	February–December 2022	Sole traders' assets separated; allowance.
Business creation statistics	INSEE	February 2022 Statistics	2021	Record 995,900 creations in 2021.
Composition of the Government	President	20 May 2022 Decree	May 2022	Borne government; ministry renamed.
Purchasing-power Act	Parliament	16 August 2022 Act	July–December 2022	Value-sharing bonus exempt up to EUR 3,000.
Draft finance bill for 2023	MEFIDS	26 September 2022 Press kit	September 2022	Production-tax abolition proposed.
Bpifrance, first half of 2022	Bpifrance	27 September 2022 Press release	January–June 2022	Innovation capital EUR 415 million; honour loans.
French Tech Next40/120 call	MEFIDS	20 October 2022 Press release	October 2022	Fundraising threshold EUR 40 million.
One-stop shop for formalities	MEFIDS	16 November 2022 Press release	November 2022	Online from November; compulsory in 2023.
Senate opinion on the Economy mission	Senate	16 November 2022 Committee opinion	November 2022	Draft cuts to crafts and commerce funds.
Tech IPO measures	MEFIDS	23 November 2022 Press release	November 2022	Listing goal; public IPO money; voting rights.
First Factory call	MEFIDS	24 November 2022 Press kit	January–November 2022	EUR 550 million call; 18 first winners.
University innovation hubs	MEFIDS	7 December 2022 Press release	December 2022	Call for 20 more hubs; EUR 160 million.
Finance Act for 2023	Parliament	30 December 2022 Act	December 2022	Production-tax rates halved for 2023.
Germany				
Budget for 2022, section 09	Budget Committee	2 May 2022 Recommendation	2022	Start-up title and INVEST raised.
KfW Entrepreneurship Monitor	KfW	May 2022 Report	2021	607,000 founders in 2021; discharge little known.

Subject	Issuing bodies	Date and type	Period covered	Description
Future Financing Act key points	Federal Ministry of Justice	29 June 2022 Press release	June 2022	Multiple votes; listing capital; employee shares.
Start-up Strategy	Federal Government	28 July 2022 Strategy	July 2022	10 fields of action; no money of its own.
Online company formation	Federal Ministry of Justice	29 July 2022 Press release	August 2022	Online formation from 1 August.
Venture Tech Growth Financing	KfW	1 August 2022 Press release	August 2022	EUR 1.2 billion of growth loans with private lenders.
Draft federal budget for 2023	Federal Government	5 August 2022 Budget bill	2021–2022	Start-up title EUR 169.9 million.
KfW start-up report	KfW	28 September 2022 Press release	2021	61,000 start-ups; 12 % sought venture capital.
Budget committee report for 2023	Budget Committee	18 November 2022 Report	November 2022	Skills title raised; start-up title raised for a Schwedt laboratory.
Skilled immigration key points	Federal Government	30 November 2022 Key points	November 2022	Opportunity card; bills planned.
Mittelstand innovation report	Federal Government	6 December 2022 Report	2021–2022	EXIST, INVEST, Future Fund, ZIM.
Bundesrat session of 16 December	Bundesrat	8 December 2022 Record	November–December 2022	Budget Act 2023 passed by the Bundestag.

Source: The plans, laws, reports and announcements listed. Periods covered and descriptions: IFEED.

Note: Where a source has several parts, citations name the part (for example, work plan or policy note) before the page. Abbreviations: BEIS, Department for Business, Energy and Industrial Strategy; CBIRC, China Banking and Insurance Regulatory Commission; CHIPS, Creating Helpful Incentives to Produce Semiconductors; CP, Command Paper; CSOP, Company Share Option Plan; DBJ, Development Bank of Japan; EU, European Union; EXIST, Business Start-ups from Science (federal programme); FILP, Fiscal Investment and Loan Program; FinCEN, Financial Crimes Enforcement Network; FSA, Financial Services Agency; FY, fiscal year; GmbH, limited company under German law; HC, House of Commons paper; INSEE, National Institute of Statistics and Economic Studies; INVEST, grant for venture capital (federal programme); IPO, initial public offering; JFC, Japan Finance Corporation; JST, Japan Science and Technology Agency; M&A, mergers and acquisitions; MEFIDS, Ministry of the Economy, Finance and Industrial and Digital Sovereignty; METI, Ministry of Economy, Trade and Industry; MIIT, Ministry of Industry and Information Technology; MOEF, Ministry of Economy and Finance; MOF, Ministry of Finance (China); MOST, Ministry of Science and Technology; MSS, Ministry of SMEs and Startups; NDRC, National Development and Reform Commission; NEDO, New Energy and Industrial Technology Development Organization; NSF, National Science Foundation; R&D, research and development; SAMR, State Administration for Market Regulation; SBA, U.S. Small Business Administration; SBDC, Small Business Development Center; SBIC, Small Business Investment Company; SBIR, Small Business Innovation Research; SBLC, Small Business Lending Company; SEIS, Seed Enterprise Investment Scheme; SME, small and medium-sized enterprise; SRDI, specialised, refined, distinctive and innovative; SSBCI, State Small Business Credit Initiative; STA, State Taxation Administration; STTR, Small Business Technology Transfer; TIPS, Tech Incubator Program for Startup; UK, United Kingdom; US (U.S.), United States; VAT, value added tax; ZIM, Central Innovation Programme for SMEs (federal programme).

2 REVIEW OF ENTREPRENEURIAL ECOSYSTEM POLICIES

This section reviews the key policies of 2022 in six domains adapted from those of Isenberg (2011): regulation and law; finance; culture; institutional support; human capital, in education, training and labour; and markets, including initial public offerings (IPO) and exit. One standard is applied throughout: a policy is good where every measure judged was in force, launched or enacted by the end of the year and reaches the firms it is meant for; mixed where such measures sit beside pending, planned or announced ones, reach other targets, or are all pending but aimed at founders; and poor where nothing was in force and the measures were aimed away from founders; events, awards and proclamations do not count as measures in force.

2.1 REGULATION AND LAW

Regulation turned on formation, registers and exit.

2.1.1 KOREA: THE START-UP SUPPORT ACT AND THE VENTURE INVESTMENT DECREE

The revised Act on Support for Small and Medium Business Start-ups (the Start-up Support Act) took full effect on 29 June 2022; its decree, approved by the Cabinet on 21 June, raised from 30 % to more than 50 % the share that an existing business may hold in a new company before the company ceases to count as a start-up (MSS, 21 June 2022, pp. 1–2). On 16 August the Cabinet approved a decree of the Venture Investment Act, due to take effect on 23 August, that halves to KRW 1 billion the minimum size of an accelerator's venture partnership and limits the joint liability of an investee's officers and largest shareholders (MSS, 16 August 2022, pp. 1, 4). Multiple-voting shares for unlisted venture firms remained a bill pending before the National Assembly (MSS, 12 July 2022, p. 5).

This was a mixed policy. The wider definition lets experienced founders and investing companies take stakes in new firms without the firms losing start-up support, and the decree eases accelerator funds. But multiple-voting shares, which would let founders raise large rounds without losing control, remained a bill.

2.1.2 JAPAN: THE BUSINESS-GROWTH SECURITY RIGHT

On 2 November the FSA put to a working group of its Financial System Council a security right over a whole business, intangible assets included, to make founding easier; its paper noted that about half of those who borrowed from private banks at founding had given a personal guarantee (FSA, 2 November 2022, pp. 5, 26). The Start-up Development Five-year Plan of 28 November undertook to submit a bill "early" (Council for the Realisation of New Capitalism, 28 November 2022, p. 18), and the Management Guarantee Reform Programme of 23 December announced that from April 2023 lenders will have to explain to borrowers why a personal guarantee is needed (METI, FSA, & Ministry of Finance of Japan, 23 December 2022, p. 2).

This was a mixed policy. It has been argued that a lower personal cost of failure encourages risky entry (Lee et al., 2007), and a security over the whole business would let founders borrow on its

value rather than on personal guarantees. But the security right was a study and a promised bill, and the stricter supervision an announcement.

2.1.3 CHINA: SAMR ORDER NO. 52 AND STATE COUNCIL DECREE NO. 755

SAMR Order No. 52, in force on its promulgation on 1 March, gives electronic business licences the force of paper ones, lets a market entity hit by a disaster or public-health event suspend business for up to three years, and provides simplified deregistration with the commitment of all investors and 20 days' public notice (SAMR, 1 March 2022, Arts. 23, 40–42, 44, 47). State Council Decree No. 755, signed by Premier Li Keqiang and in force from 1 November, provides for equal market access for individual businesses and lets them convert into companies, with simpler annual reports (State Council of the People's Republic of China, 1 October 2022, preamble, Arts. 5, 6, 12, 13).

This was a good policy. Costly entry regulation hampers the entry of new firms (Klapper et al., 2006), and the rules, in force, lower the cost of entering, pausing and leaving a business; it has been argued that a lower cost of failure encourages risky entry (Lee et al., 2007).

2.1.4 UNITED KINGDOM: THE REGISTER REFORM WHITE PAPER AND THE GROWTH PLAN

Under Boris Johnson's government the white paper of 28 February on register reform proposed that directors, controllers and filers have a verified identity and that the registrar be able to reject filings, while undertaking to "maintain our user-friendly, low-cost framework" (BEIS, 28 February 2022, pp. 8, 13). Liz Truss's government made "cutting red tape" a priority of its Growth Plan of 23 September and announced a legislative programme of supply-side reforms (HM Treasury, 23 September 2022, para. 3.6). Rishi Sunak's government stated on 17 November that it had "reversed nearly all the measures in the Growth Plan" and undertook to review retained EU law (HM Treasury, 17 November 2022, p. 1, and para. 5.69).

This was a poor policy. Costly entry regulation hampers the entry of new firms (Klapper et al., 2006), so the undertaking to keep incorporation cheap matters. But the register reform, which adds checks for all companies, awaited a bill, and the deregulatory agenda was a general programme and a review.

2.1.5 UNITED STATES: THE SBA BURDEN-REDUCTION RULE AND BENEFICIAL OWNERSHIP REPORTING

A final rule of the SBA, effective from 1 August 2022, removed or revised obsolete or burdensome rules of its 7(a), Microloan and 504 loan programmes (SBA, 30 June 2022, p. 38900). On 26 October the SBA proposed to base affiliation in its business loan programmes on ownership alone and to require hazard insurance only on loans above USD 150,000 (SBA, 26 October 2022, pp. 64724–64726). FinCEN's final rule of 30 September will require reporting companies to identify their beneficial owners from 1 January of a later year; FinCEN estimated about 32.6 million reporting companies in the first year, and commenters raised the burden on small businesses (FinCEN, 30 September 2022, pp. 59498, 59509, 59549).

This was a mixed policy. The burden-reduction rule was in force, and the affiliation proposal would widen access to guaranteed loans. But the proposal was not final, and costly entry regulation hampers the entry of new firms (Klapper et al., 2006), while ownership reporting will add a duty for millions of companies, new ones among them.

2.1.6 FRANCE: THE SOLE-TRADER STATUTE AND THE ONE-STOP SHOP

Act No. 2022-172 of 14 February, signed under Jean Castex's government, separates a sole trader's professional assets from personal ones, so that business creditors can in general pursue only the professional assets, from three months after promulgation (Parliament of France, 14 February 2022, Arts. 1, 19). The online one-stop shop for business formalities opened in November for creations, changes and cessations; it replaces the networks of formalities centres and 56 paper forms, and its use will be compulsory from 1 January 2023 (MEFIDS, 16 November 2022, pp. 1–2).

This was a good policy. It has been argued that a lower personal cost of failure encourages risky entry (Lee et al., 2007), and the separation of assets limits what a failed sole trader can lose; costly entry regulation hampers entry (Klapper et al., 2006), and a single online channel lowers the cost of founding. Both measures were in force or launched.

2.1.7 GERMANY: ONLINE GMBH FORMATION AND THE FUTURE FINANCING ACT KEY POINTS

From 1 August 2022 a limited company (GmbH) can be formed online with cash contributions, by video with a notary, and register extracts became free; formations with contributions in kind will follow from 1 August 2023 (Federal Ministry of Justice, 29 July 2022, press release). The key points of the justice and finance ministries of 29 June for a Future Financing Act would allow multiple-voting shares and cut the minimum capital for a listing from EUR 1.25 million to EUR 1 million (Federal Ministry of Justice, 29 June 2022, key points), and the Start-up Strategy announced an act on regulatory sandboxes and the aim of formation within 24 hours (Federal Government of Germany, 28 July 2022, pp. 8, 15).

This was a mixed policy. Online formation lowers a cost of entry (Klapper et al., 2006). But the capital-market reforms and the sandbox act were key points and announcements, and formation within 24 hours an aim.

2.2 FINANCE: GRANTS, FUNDING AND FINANCIAL INFRASTRUCTURE

Finance was the largest domain: funds of funds, guarantees and tax reliefs.

2.2.1 KOREA: THE FUND OF FUNDS AND THE PRIVATE FUNDS OF FUNDS

Under Moon Jae-in's government the Fund of Funds call of 22 February offered KRW 929.7 billion from ministries and agencies to form at least KRW 1.6 trillion of venture funds, including a new KRW 40 billion regional angel fund (MSS, 22 February 2022, pp. 3–4). Yoon Suk Yeol's government proposed on 21 July to extend tax breaks for start-up and venture investment by three years (MOEF, 21 July 2022, summary, p. 14). On 4 November the ministry reported third-quarter venture

investment KRW 838.8 billion below a year earlier, after rises in the first two quarters, and announced private funds of funds with tax incentives (MSS, 4 November 2022, pp. 1, 4). The budget proposal for 2023 put KRW 313.5 billion into the Fund of Funds (MSS, 31 August 2022, p. 3).

This was a mixed policy. The Fund of Funds commits public money to funds whose private managers choose the firms, and public money that invests alongside private investors is associated with more investment and more successful exits (Brander et al., 2015). But the private funds of funds and the investor tax breaks were an announcement and a proposal.

2.2.2 JAPAN: THE GUARANTEE WITHOUT A PERSONAL GUARANTEE AND THE REINVESTMENT RELIEF

The Management Guarantee Reform Programme of 23 December announced a credit guarantee without a personal guarantee for founders and firms less than five years old, a 100 % guarantee up to JPY 35 million, to start in March 2023; Shoko Chukin Bank had in principle abolished management guarantees on its start-up loans from October 2022 (METI, FSA, & Ministry of Finance of Japan, 23 December 2022, p. 1). The five-year plan put JPY 12 billion towards the guarantee corporations' losses (Council for the Realisation of New Capitalism, 28 November 2022, p. 15), and the FILP plan of 23 December doubled to JPY 1.44 billion the ceiling of a JFC loan programme for start-ups (Ministry of Finance of Japan, 23 December 2022, p. 1). The tax outline of 23 December, a Cabinet decision to be enacted in 2023, would exempt gains on shares, up to JPY 2 billion, that individuals put into founding a new company (Government of Japan, 23 December 2022, p. 5).

This was a mixed policy. It has been argued that a lower personal cost of failure encourages risky entry (Lee et al., 2007), and most guaranteed loans in Canada were found to be incremental (Riding et al., 2007); a larger exclusion of capital gains on start-up stock raised the funding of eligible firms (Edwards & Todtenhaupt, 2020). But only the state bank's practice was in force; the guarantee, the loan ceiling and the relief were announced, planned or proposed.

2.2.3 CHINA: TAX REFUNDS, RE-GUARANTEES AND THE VENTURE AND ANGEL TAX CONDITIONS

The government work report of 5 March expected tax refunds and cuts of about CNY 2.5 trillion (State Council of the People's Republic of China, 5 March 2022, Part III (2)); the State Council's package of 24 May raised new refunds of value added tax (VAT) credits to about CNY 1.64 trillion and added more than CNY 1 trillion of re-guarantees by the National Financing Guarantee Fund (State Council of the People's Republic of China, 24 May 2022, items 1, 4). On 9 February the MOF and the State Taxation Administration continued to 31 December 2023 the conditions for tax relief to venture capital firms and angel investors that invest in start-up technology firms with up to 300 employees and CNY 50 million each of assets and sales (MOF & STA, 9 February 2022, paras. 1–2). The CBIRC again required banks to raise inclusive loans to small firms at least as fast as all loans (CBIRC, 6 April 2022, item (2)).

This was a mixed policy. The refunds, re-guarantees and tax conditions were in force and reached small firms at scale, and the venture and angel conditions give investors a stable test. But most of

the money served relief for all small firms rather than founders' risk capital, and credit was steered by volume targets; entrepreneurship policy has been urged to show the market failures it addresses rather than assume them (Acs, Åstebro, Audretsch, & Robinson, 2016).

2.2.4 UNITED KINGDOM: START UP LOANS, THE SEED SCHEME AND THE SME R&D RELIEF

Under Boris Johnson's government the Finance Act 2022 extended the GBP 1 million annual investment allowance to 31 March 2023 (Parliament of the United Kingdom, 24 February 2022, s. 12), and on 20 July the Recovery Loan Scheme was extended for two years, but lenders "may now require a personal guarantee" (BEIS, 20 July 2022, press release). Under Liz Truss's government Start Up Loans were widened on 25 September to firms trading for up to three years (BEIS, 25 September 2022, press release), and the Growth Plan announced from April 2023 higher limits for the Seed Enterprise Investment Scheme (SEIS), of GBP 250,000 per company and GBP 200,000 per investor (HM Treasury, 23 September 2022, para. 4.7). Rishi Sunak's government kept them and announced that, from 1 April 2023, the additional deduction for research and development (R&D) by SMEs will fall from 130 % to 86 % and their credit from 14.5 % to 10 %, while the credit for larger firms rises from 13 % to 20 % (HM Treasury, 17 November 2022, paras. 3.31, 5.52).

This was a mixed policy. Start Up Loans reach young firms, the enacted allowance reaches any firm investing in plant and machinery, and relief on start-up shares has raised the funding of eligible firms (Edwards & Todtenhaupt, 2020). But the SEIS limits were announced, not enacted, the Recovery Loan Scheme let lenders require personal guarantees, which it has been argued raise the personal cost of failure (Lee et al., 2007), and relief for R&D by SMEs was to be cut.

2.2.5 UNITED STATES: THE SBIC AND SBLC PROPOSALS AND THE RESEARCH CREDIT

The SBA proposed on 19 October to revise its rules for Small Business Investment Companies (SBIC), with an Accrual Debenture for funds that put at least 75 % of their financings into equity and lower licensing fees for new funds (SBA, 19 October 2022, pp. 63436, 63439, 63441), and on 7 November to lift the moratorium on new Small Business Lending Companies (SBLC) (SBA, 7 November 2022, p. 66963); neither rule was final. Community Advantage pilot changes took effect on 31 May (SBA, 29 April 2022, p. 25398), and the President said the State Small Business Credit Initiative (SSBCI) is helping entrepreneurs tap USD 10 billion in investment and loans (President of the United States, 31 October 2022, p. 66525). The Inflation Reduction Act raises by USD 250,000 the research credit that qualified small businesses may set against payroll tax, for taxable years beginning after 31 December 2022 (U.S. Congress, 16 August 2022, sec. 13902).

This was a mixed policy. The Accrual Debenture would put public leverage behind privately managed equity funds, as the evidence favours (Brander et al., 2015), and the pilot changes and the payroll offset reach small firms. But the SBIC and SBLC reforms were proposals.

2.2.6 FRANCE: THE PRODUCTION-TAX CUT AND THE FIRST FACTORY CALL

The draft finance bill of 26 September proposed abolishing the contribution on firms' value added over two years (MEFIDS, 26 September 2022, p. 10); the Finance Act for 2023 of 30 December

halves its rates for 2023, the top rate from 0.75 % to 0.375 %, and applies most of the abolition from the following year (Parliament of France, 30 December 2022, Art. 55). Under the French investment plan the industrial start-ups strategy has EUR 2.3 billion and its First Factory call EUR 550 million; on 24 November 18 projects were chosen from nearly 100 applications, for EUR 88 million of aid (MEFIDS, 24 November 2022, pp. 10, 12). In the first half Bpifrance invested EUR 415 million in innovation capital, a rise it gives as 38 % in its summary and 37 % in its tables, while state-guaranteed loans fell to EUR 2.85 billion from EUR 9 billion (Bpifrance, 27 September 2022, pp. 1, 2, 4, 8).

This was a mixed policy. The halving of the levy is enacted, and Bpifrance's innovation capital rose. But the First Factory call awards aid to projects chosen by the state, whereas it has been argued that high-growth firms are too varied to be identified in advance (Mason & Brown, 2013).

2.2.7 GERMANY: THE FUTURE FUND MODULES AND THE INVEST GRANT

The Future Fund, a EUR 10 billion federal fund at KfW (Federal Government of Germany, 28 July 2022, p. 4), added modules in 2022: Venture Tech Growth Financing 2.0, launched on 1 August with EUR 1.2 billion to lend EUR 1 million to EUR 125 million to growth firms alongside private lenders, who must generally provide half (KfW, 1 August 2022, press release), and the DeepTech & Climate Fund, of up to EUR 1 billion, investing from September (Federal Government of Germany, 6 December 2022, p. 3). The Budget Committee raised INVEST, the grant for business angels, from EUR 44.2 million to EUR 53.1 million for 2022 (Budget Committee of the German Bundestag, 2 May 2022, p. 4). The draft budget for 2023 put the title for innovative start-ups at EUR 169.9 million, against EUR 179.9 million in 2022, with INVEST at EUR 45.9 million (Federal Government of Germany, 5 August 2022, ch. 0902, title 686 07); the committee's increase was earmarked for a start-up laboratory in Schwedt, and the Bundestag passed the budget on 25 November (Budget Committee of the German Bundestag, 18 November 2022, p. 33; Bundesrat, 8 December 2022, p. 26).

This was a mixed policy. The fund modules lend and invest alongside private investors, as evidence on public co-investment favours (Brander et al., 2015), and INVEST lowers a first outside investor's cost. But the draft for 2023 cut INVEST, and the committee's rise went to one laboratory.

2.3 CULTURE

Culture remained the thinnest domain.

2.3.1 KOREA: THE RESTARTER INTEGRITY EVALUATION

On 16 March, under Moon Jae-in's government, the ministry relaunched the integrity evaluation that restarting entrepreneurs must pass to receive restart support: one designated body replaced several, applications became possible all year and the items were cut from five to three; from 2017 to 2021, 11,244 people had applied and 10,461 passed (MSS, 15 March 2022, pp. 1–2). Under Yoon Suk Yeol's government the ministry held Re-challenge Day on 22 November to spread a culture of re-challenge after failure, honouring 10 businesspeople (MSS, 22 November 2022, p. 1), and its work

plan stated that Korea's rank by number of unicorns had fallen from sixth to 10th in three years, while their number rose from 11 to 22 (MSS, 27 December 2022, work plan, p. 3).

This was a mixed policy. It has been argued that stigma can blunt the effect of rules friendly to entrepreneurs (Lee et al., 2007), and a simpler test eases the return of failed founders to support. But the other measure on attitudes to failure was an event.

2.3.2 JAPAN: THE RE-CHALLENGE RULE AND THE FIRST YEAR OF START-UPS

The five-year plan found that 25 % of Japanese see founding a business as a desirable career, against 79 % in China and 68 % in the United States, and that its business opening rate, 5.1 %, trails those of the United States (9.2 %) and the United Kingdom (11.9 %) (Council for the Realisation of New Capitalism, 28 November 2022, pp. 2, 4). The economic measures of 28 October named 2022 "the first year of start-ups" (Government of Japan, 28 October 2022, p. 34). For re-challenge, the plan records that the Employment Insurance Act was amended in 2022 so that a founder may exclude up to three years of business from the period for claiming unemployment benefit; it gives no date of effect (Council for the Realisation of New Capitalism, 28 November 2022, p. 9).

This was a mixed policy. Tolerance of risk and failure belongs to the culture of an entrepreneurial ecosystem (Isenberg, 2011), and the enacted rule lowers the cost of a failed attempt. But the government otherwise answered a measured problem with a declared first year and with finance (Section 2.2.2).

2.3.3 CHINA: THE WORK REPORT PLEDGE AND THE MASS INNOVATION WEEK

The government work report pledged to "promote the entrepreneurial spirit" (State Council of the People's Republic of China, 5 March 2022, Part III (3)). Decree No. 755 asks governments to commend outstanding individual businesses (State Council of the People's Republic of China, 1 October 2022, Art. 29), and the NDRC and other departments undertook to plan and prepare the national mass entrepreneurship and innovation week of 2022 to build a positive atmosphere for graduates' entrepreneurship (NDRC, 8 February 2022, Part 5, item (1)).

This was a poor policy. Tolerance of failure belongs to the culture of an entrepreneurial ecosystem (Isenberg, 2011); the government answered it with a pledge, commendations and a planned week of events.

2.3.4 UNITED KINGDOM: THE QUEEN'S AWARDS FOR ENTERPRISE AND THE MANSION HOUSE SPEECH

On 21 April 232 businesses received the Queen's Awards for Enterprise, 141 of them for international trade and 51 for innovation (BEIS, 21 April 2022, press release). At Mansion House in July the Chancellor, Nadhim Zahawi, said that he knew "the value of entrepreneurship, start-ups, and innovation" and that growth comes "from the wealth creators and risk takers" (HM Treasury, 20 July 2022, speech).

This was a poor policy. Tolerance of failure belongs to the culture of an entrepreneurial ecosystem (Isenberg, 2011); Boris Johnson's government answered it with awards and a speech.

2.3.5 UNITED STATES: NATIONAL SMALL BUSINESS WEEK AND NATIONAL ENTREPRENEURSHIP MONTH

The President proclaimed National Small Business Week from 1 to 7 May (President of the United States, 29 April 2022, p. 26668) and November 2022 National Entrepreneurship Month, stating that "a record 5.4 million new businesses were started in 2021" (President of the United States, 31 October 2022, pp. 66525–66526).

This was a poor policy. The second proclamation counted as businesses what the first had counted as applications, and it has been argued that encouraging more people to start firms is not by itself good public policy (Shane, 2009).

2.3.6 FRANCE: HONOUR LOANS AND THE ALLOWANCE AFTER A NON-VIABLE BUSINESS

INSEE counted a record 995,900 business creations in 2021, including 641,500 micro-entrepreneurs (INSEE, February 2022, p. 1). In the first half of 2022 Bpifrance disbursed nearly 7,400 honour loans worth EUR 53 million (Bpifrance, 27 September 2022, p. 11). Article 11 of the Act of 14 February extends the allowance for independent workers to those whose business has ceased for good because it was not economically viable, as attested by a trusted third party, for claims from the month after its publication (Parliament of France, 14 February 2022, Arts. 11, 19).

This was a mixed policy. The allowance after a non-viable closure lowers the personal cost of failure, which it has been argued encourages risky entry (Lee et al., 2007), and honour loans bring finance to founders. But the record counts registrations, not growing firms (Shane, 2009), and no measure addressed the stigma of failure.

2.3.7 GERMANY: THE START-UP STRATEGY AND THE FOUNDING-CULTURE BUDGET LINE

The Start-up Strategy welcomed founding "from all walks of life and a culture of the second chance" (Federal Government of Germany, 28 July 2022, p. 2). The Budget Committee raised the line for strengthening founding culture and succession from EUR 4.75 million to EUR 7.6 million for 2022, and the draft for 2023 put it at EUR 5.0 million (Budget Committee of the German Bundestag, 2 May 2022, p. 4; Federal Government of Germany, 5 August 2022, ch. 0902, title 686 07); EXIST Start-up Culture supports 142 universities (Federal Government of Germany, 6 December 2022, p. 2). KfW found that only 28 % of the working population knew of the shorter discharge from residual debt (KfW, May 2022, pp. 3, 11).

This was a mixed policy. The line for 2022 and EXIST Start-up Culture fund founding culture, and bankruptcy laws friendly to entrepreneurs go with higher entry (Lee et al., 2011). But few knew of the shorter discharge, the second chance remained an aim, and the draft for 2023 cut the line.

2.4 SUPPORT: INSTITUTIONAL SUPPORT

Support turned on private operators, state selection and hubs.

2.4.1 KOREA: TIPS AND THE SUPER-GAP START-UP PROJECT

The ministry's budget for 2023, passed by the National Assembly on 23 December, raises TIPS (Tech Incubator Program for Startup), in which private operators select, incubate and invest in start-ups before public support follows, from KRW 293.5 billion to KRW 378.2 billion and to at least 700 teams, against 500 in 2022, and funds a project to select 1,000 "super-gap" start-ups in 10 new fields over five years (MSS, 26 December 2022, pp. 1–2). The K-Startup strategy of 8 September would hand COMEUP, the largest start-up event, to the private sector and open a K-Startup Center in Vietnam (MSS, 8 September 2022, pp. 2–3).

This was a mixed policy. TIPS lets private investors choose the firms that public money follows, as the evidence on co-investment favours (Brander et al., 2015). But the super-gap project selects firms in fields chosen by the state, whereas it has been argued that high-growth firms are too varied to be identified in advance (Mason & Brown, 2013), and the overseas measures were announcements.

2.4.2 JAPAN: THE JST FUND, THE NEDO FUND AND THE GLOBAL STARTUP CAMPUS

The five-year plan would create a JPY 100 billion, five-year fund at the Japan Science and Technology Agency (JST) to commercialise university research in at least 5,000 cases, and a Global Startup Campus combining deep-tech research with incubation (Council for the Realisation of New Capitalism, 28 November 2022, pp. 6–7); the Cabinet Office recorded a new JPY 100 billion fund over five years at NEDO for research-based start-ups (Cabinet Office of Japan, 23 December 2022, p. 3). The plan stated that the Cabinet had decided about JPY 1 trillion of budget measures for start-ups (Council for the Realisation of New Capitalism, 28 November 2022, p. 3).

This was a mixed policy. The funds are aimed at research-based start-ups, but it has been argued that ecosystems grow organically rather than by design (Isenberg, 2010), and the campus is a hub designed by the state; the funds and the campus were announcements in plans.

2.4.3 CHINA: THE SRDI LADDER AND NATIONAL INCUBATORS

MIIT's interim measures of 1 June define three tiers of quality SMEs, the highest being specialised, refined, distinctive and innovative (SRDI) "little giants", and set targets of 1 million, 100,000 and 10,000 firms over the 14th Five-Year Plan (MIIT, 1 June 2022, Arts. 2, 6); the State Council's package added about 500 "little giants" in 2022 (State Council of the People's Republic of China, 24 May 2022, item 16). MOST recognised 149 national-level science and technology business incubators on 19 April (MOST, 19 April 2022, para. 1), and start-up venues funded by government must give about 30 % of their space free to graduate founders (General Office of the State Council of the People's Republic of China, 5 May 2022, Part 1, item (3)).

This was a mixed policy. Incubators and free space bring support within reach of founders. But the ladder chooses firms by a state designation of status, whereas it has been argued that high-growth firms are too varied to be identified in advance (Mason & Brown, 2013).

2.4.4 UNITED KINGDOM: HELP TO GROW: MANAGEMENT AND INVESTMENT ZONES

Under Boris Johnson's government Help to Grow: Management, training with 90 % of the cost met from GBP 220 million of public funding, was widened on 14 June to two participants for firms with 10 or more employees (BEIS & HM Treasury, 14 June 2022, press release). Liz Truss's government announced Investment Zones with time-limited tax reliefs (HM Treasury, 23 September 2022, para. 4.12); Rishi Sunak's government refocused them on a few knowledge-intensive clusters and dropped the existing expressions of interest (HM Treasury, 17 November 2022, para. 3.25).

This was a mixed policy. Help to Grow reaches the leaders of small firms. But Investment Zones were refocused within two months of their announcement, and it has been argued that public programmes fail when they are too short-lived (Lerner, 2009).

2.4.5 UNITED STATES: ENTREPRENEURIAL DEVELOPMENT PROGRAMS AND THE SBDC PROPOSAL

The Consolidated Appropriations Act, 2023, of 29 December provides USD 320 million for the SBA's Entrepreneurial Development Programs, against a request of USD 318 million, including USD 140 million for Small Business Development Centers (SBDC), for which USD 136 million had been requested (U.S. Congress, 29 December 2022, 136 Stat. 4694; SBA, 2022, p. 14). On 13 December the SBA proposed the first revision of the centres' rules since 1995, with a client opt-in that would let it survey outcomes (SBA, 13 December 2022, pp. 76129, 76131), and the CHIPS and Science Act has the Department of Energy fund lab-embedded entrepreneurship programmes (U.S. Congress, 9 August 2022, sec. 10717).

This was a mixed policy. The enacted funding sustains a network of advice, and the lab-embedded programmes fund the connections that ecosystem research recommends (Brown & Mason, 2017). But the rule for the centres was a proposal.

2.4.6 FRANCE: THE FRENCH TECH PROGRAMMES AND THE UNIVERSITY INNOVATION HUBS

The call of 20 October for the French Tech Next40/120 doubled the threshold of its fundraising track to EUR 40 million raised in 2020 to 2022 and offered selected firms tailored support through French Tech correspondents in administrations (MEFIDS, 20 October 2022, p. 2). On 7 December the state opened a call for 20 more university innovation hubs with EUR 160 million (MEFIDS, 7 December 2022, p. 2), while the draft budget cut the tax resources of the chambers of crafts by EUR 15 million and abolished the fund for local commerce and crafts, as the Senate's Committee on Economic Affairs reported (Senate of France, 16 November 2022, p. 5).

This was a mixed policy. The hubs build connections between research and start-ups, as ecosystem research recommends (Brown & Mason, 2017). But the Next40/120 concentrates

support on firms that have already raised large sums, whereas it has been argued that policy should support the conditions for growth rather than select firms (Mason & Brown, 2013), and the draft budget cut support for crafts and local commerce.

2.4.7 GERMANY: THE START-UP STRATEGY, EXIST AND ZIM

The Start-up Strategy of July sets out 10 fields of action, to be financed within existing budget ceilings and reported on every year (Federal Government of Germany, 28 July 2022, p. 3), and announced an EXIST Women line and contact points for start-ups in every federal ministry (Federal Government of Germany, 28 July 2022, pp. 10, 16). EXIST, for start-ups from science, approved 227 founder grants in 2021, and the stop on applications to ZIM, in place since October 2021, was lifted in early August 2022 (Federal Government of Germany, 6 December 2022, pp. 2, 9).

This was a mixed policy. EXIST and ZIM fund founders and innovative SMEs, and ZIM reopened. But a programme closed to applications for 10 months is the kind of instability that it has been argued makes public programmes fail (Lerner, 2009), and the strategy's new lines were announcements without money of their own.

2.5 HUMAN CAPITAL: EDUCATION AND TRAINING, AND LABOUR

Policy on people turned on stock options, visas and graduates.

2.5.1 KOREA: STOCK OPTIONS AND THE TECHNOLOGY START-UP VISA

The tax-free ceiling on gains from exercising stock options had risen from KRW 30 million to KRW 50 million a year from 2022, and a decree approved on 15 February, with no date of effect stated, would let venture firms value unlisted shares for options by more methods (MSS, 15 February 2022, pp. 1, 3). The tax revision proposal of 21 July would raise the ceiling to KRW 200 million (MOEF, 21 July 2022, summary, p. 15), and the amendment of the Venture Business Act approved by the Cabinet on 27 December would cap options for outside experts at 10 % of shares and oblige firms to report grants (MSS, 27 December 2022, amendment release, pp. 2–3). The K-Startup strategy would make the renewal of the technology start-up visa in two-year units practicable (MSS, 8 September 2022, p. 3).

This was a mixed policy. The higher ceiling lets firms short of cash pay key staff in equity, and the wider valuation methods would. But immigrants are more likely than natives with similar education to start companies (Hunt, 2011), and the visa change was an announcement, while the further rise in the ceiling and the rules for experts were a proposal and an amendment not yet in force.

2.5.2 JAPAN: STOCK-OPTION PERIODS AND THE START-UP VISA PLAN

The tax outline of 23 December would lengthen from 10 to 15 years the exercise period of tax-qualified stock options granted by unlisted companies less than five years old (Government of Japan, 23 December 2022, p. 8). The five-year plan would let certified venture capital firms and accelerators, not only local governments, confirm start-up visa applications and would lengthen

the stay, at present at most one year (Cabinet Office of Japan, 23 December 2022, p. 3; Council for the Realisation of New Capitalism, 28 November 2022, p. 9).

This was a mixed policy. Longer exercise periods would let young firms pay early employees in equity, and private confirmation would widen a narrow route for foreign founders, who are more likely than natives with similar education to start companies (Hunt, 2011). But the stock-option and visa changes were a tax outline and plans.

2.5.3 CHINA: THE NOTICE ON GRADUATES' EMPLOYMENT AND ENTREPRENEURSHIP

The notice of the General Office of the State Council of 5 May gives SMEs that hire graduates of the year a one-off subsidy, valid to the end of 2022, and gives graduate founders a start-up subsidy, guaranteed loans with interest subsidies and tax relief, while universities are to deepen entrepreneurship education (General Office of the State Council of the People's Republic of China, 5 May 2022, Part 1, items (1), (3)). The NDRC's notice of 8 February directed demonstration bases to help about 10.76 million graduates start businesses and find jobs (NDRC, 8 February 2022, Part 2, item (1)).

This was a mixed policy. The measures were in force and support graduate founders. But they address youth employment more than entrepreneurship, and an evaluation of a leading entrepreneurship programme for students found a negative effect on the intention to become an entrepreneur (Oosterbeek, van Praag, & Ijsselstein, 2010).

2.5.4 UNITED KINGDOM: THE SCALE-UP AND HIGH POTENTIAL INDIVIDUAL ROUTES

Under Boris Johnson's government the immigration rules laid on 15 March (HC 1118) provided for a High Potential Individual route for recent graduates of top universities abroad and a Scale-up route for holders of a job offer of at least GBP 33,000 from a firm that grew by 20 % a year for three years and had at least 10 employees at the start; after six months Scale-up holders are no longer tied to the sponsor (Home Office, 15 March 2022, paras. 7.13–7.21). Liz Truss's government promised an immigration plan for growth "in the coming weeks" and announced that the Company Share Option Plan (CSOP) limit would double to GBP 60,000 from April 2023 (HM Treasury, 23 September 2022, paras. 3.27, 4.7), which Rishi Sunak's government kept (HM Treasury, 17 November 2022, para. 3.31).

This was a mixed policy. The routes let graduates and the staff of growing firms come from abroad, and immigrants are more likely than natives with similar education to start companies (Hunt, 2011). But the immigration plan was a promise and the CSOP increase an announcement.

2.5.5 UNITED STATES: ENTREPRENEURIAL FELLOWSHIPS AND LEAVE IN THE CHIPS AND SCIENCE ACT

The CHIPS and Science Act requires NSF to award entrepreneurial fellowships to early-career scientists and engineers, and lets national laboratory staff take leave to commercialise energy technologies with the option to return up to three years later (U.S. Congress, 9 August 2022, secs. 10392, 10719). The SBA requested USD 4.5 million for entrepreneurship education, against USD 2.5 million under the continuing resolution for 2022 (SBA, 2022, p. 14).

This was a mixed policy. The enacted leave and fellowships lower the career cost of founding. But they reach scientists, not founders at large, and the education line was a request.

2.5.6 FRANCE: THE VALUE-SHARING BONUS AND THE HUBS' ENTREPRENEURSHIP TRAINING

The Act of 16 August on emergency measures to protect purchasing power lets employers pay a value-sharing bonus exempt up to EUR 3,000 a year per beneficiary, or EUR 6,000 where the firm has a profit-sharing or incentive scheme (Parliament of France, 16 August 2022, Art. 1). The university innovation hubs called for on 7 December are to offer training in entrepreneurship alongside support for deep-tech start-ups (MEFIDS, 7 December 2022, p. 2).

This was a mixed policy. The hubs tie entrepreneurship training to support for start-ups. But the bonus is cash, not equity, and reaches the employees of all firms.

2.5.7 GERMANY: THE EMPLOYEE-SHARE PROPOSAL AND THE SKILLED-IMMIGRATION KEY POINTS

The key points of 29 June would raise the tax-free allowance for employee shares from EUR 1,440 to EUR 5,000 a year and widen deferred taxation (Federal Ministry of Justice, 29 June 2022, key points). The cabinet's key points of 30 November on skilled immigration would add an opportunity card for job search and lower the salary threshold for specialists in information technology, with bills in the first quarter of 2023 (Federal Government of Germany, 30 November 2022, pp. 1–2, 5–6). The Budget Committee raised skills-securing for SMEs to EUR 25.0 million for 2022; the draft for 2023 lowered it to EUR 22.5 million, and the committee raised it again (Budget Committee of the German Bundestag, 2 May 2022, p. 4; Budget Committee of the German Bundestag, 18 November 2022, p. 33; Federal Government of Germany, 5 August 2022, ch. 0902, title 686 05).

This was a mixed policy. The skills line supports training in small firms. But the higher allowance and the immigration reform were key points, and these make no provision for founders, although first-generation immigrants create about a quarter of new US firms (Kerr & Kerr, 2020).

2.6 MARKET: MARKET AND INDUSTRY SUPPORT, IPO AND EXIT

Markets turned on public purchasing, listing rules and routes to exit.

2.6.1 KOREA: INNOVATION-PRODUCT TRIALS AND THE RULES ON ACQUISITIONS

On 28 December the Public Procurement Service, acting as first buyer, selected central, local and public bodies to trial 58 innovation products worth KRW 15.6 billion, to be supplied from January 2023 (Public Procurement Service, 28 December 2022, main text). The decree due to take effect on 23 August lets venture funds keep for five years their shares in an investee that joins a large business group through an acquisition (MSS, 16 August 2022, pp. 1, 4), and the plan of 4 November would relax the 20 % limit on investment in listed companies by venture funds for mergers and acquisitions (MSS, 4 November 2022, p. 3).

This was a mixed policy. It has been argued that procurement can create early markets for innovative firms (Edler & Georghiou, 2007), as the trial purchases do, and the decree eases exit through acquisitions. But the relaxation of the fund limit was an announcement.

2.6.2 JAPAN: THE PROCUREMENT TARGET AND THE OPEN-INNOVATION TAX RELIEF

The five-year plan found that contracts with SMEs less than 10 years old were about 1 % of government contracts, JPY 77.7 billion in fiscal year 2020, and set a target of at least 3 % "as soon as possible" (Council for the Realisation of New Capitalism, 28 November 2022, p. 14). It would let proprietary trading systems handle unlisted shares for professional investors from a cabinet order within fiscal year 2023 (Council for the Realisation of New Capitalism, 28 November 2022, p. 16). The tax outline would extend the open-innovation tax relief to purchases of existing shares that give a majority of the votes, at an estimated cost of JPY 3 billion a year (Government of Japan, 23 December 2022, pp. 36, 110).

This was a mixed policy. It has been argued that procurement can create early markets for innovative firms (Edler & Georghiou, 2007), and the government measured its share against a target. But the target, the trading rule and the tax relief were plans and a tax outline.

2.6.3 CHINA: PROCUREMENT PREFERENCES FOR SMES AND THE HAND-IN-HAND ACTION

From 1 July the MOF raised the price preference for small and micro firms in government procurement from 6–10 % to 10–20 % and, for the second half of 2022, the share of construction projects above CNY 4 million reserved for SMEs from at least 30 % to at least 40 % (MOF, 30 May 2022, sections 2–4). The Hand-in-Hand Action of MIIT and other departments of 12 May provides for matching events between large firms and SRDI SMEs and bars large firms from abusing their market power to set unreasonable payment terms (MIIT, 12 May 2022, Part 2, items (3), (5)). The work report undertook to "fully implement" the registration system for share issues (State Council of the People's Republic of China, 5 March 2022, Part III (3)).

This was a mixed policy. The preferences were in force, and firms with easier access to finance have been found to extend trade credit to those with less (Petersen & Rajan, 1997), a flow that the bar on unreasonable payment terms protects. But the higher reserved share was temporary, and the full registration system a commitment.

2.6.4 UNITED KINGDOM: THE PROCUREMENT BILL AND THE PROSPECTUS REFORM

The Procurement Bill, explained by the Cabinet Office on 12 May under Boris Johnson's government, would require contracting authorities to consider SMEs, extend 30-day payment terms and let buyers reserve competitions below thresholds for SMEs (Cabinet Office, 12 May 2022, main benefits); it remained a bill. Under Rishi Sunak's government the Edinburgh Reforms of 9 December set out a policy note under which the EUR 8 million prospectus threshold would go and offers of any size could use a regulated public offer platform; the Financial Services and Markets Bill that would allow it was expected to receive Royal Assent by spring 2023 (HM Treasury, 9 December 2022, policy note, paras. 1.10–1.11, and press release).

This was a mixed policy. Firms with easier access to finance have been found to extend trade credit to those with less (Petersen & Rajan, 1997), and 30-day public payment terms would stop the state drawing such credit from small suppliers, while IPOs on lightly regulated second markets for small companies have performed far worse in the long run (Vismara, Paleari, & Ritter, 2012), a caution for lighter offers by small firms. But the procurement reform was a bill and the prospectus reform an illustrative draft.

2.6.5 UNITED STATES: THE PROCUREMENT SCORECARD AND THE SBIR AND STTR EXTENSION

In fiscal year 2021 small businesses received a record USD 154.2 billion of federal prime contracts, 27.23 % of eligible dollars, but the number of small business vendors fell by almost 5.8 % (SBA, July 2022, p. 1). The SBIR and STTR Extension Act of 30 September extended the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programmes for three more years, added due diligence on foreign risks and required open-topic innovation activities in defence (U.S. Congress, 30 September 2022, secs. 3, 4, 7). The appropriations for 2023 provide USD 20 million for export grants to states, against a request of USD 19.5 million (U.S. Congress, 29 December 2022, 136 Stat. 4694; SBA, 2022, p. 14).

This was a mixed policy. Early-stage innovation grants can raise the chance that young firms later obtain venture capital (Howell, 2017), and the extension kept them. But fewer small firms won prime contracts.

2.6.6 FRANCE: THE TECH IPO MEASURES

On 23 November the government set the goal that 10 French unicorns list in Paris by the middle of the decade; it announced that Bpifrance planned EUR 500 million and the Caisse des Dépôts EUR 300 million to invest directly in technology listings, and that it wished to start legislative work on multiple-voting rights in listed companies so that founders keep control after listing (MEFIDS, 23 November 2022, pp. 1–2).

This was a mixed policy. Multiple-voting rights would let founders list without losing control. But the public investment was a plan and the legislative work had yet to begin.

2.6.7 GERMANY: POST-LISTING GROWTH LOANS AND THE PROCUREMENT PLATFORM

Venture Tech Growth Financing 2.0 can also lend to listed technology companies and bridge funding rounds (KfW, 1 August 2022, press release); the key points for a Future Financing Act would cut the minimum capital for a listing (Section 2.1.7); and the Start-up Strategy would have the Competence Centre for Innovative Procurement build a platform with a start-up procurement index, while an e-marketplace for public buyers was to be set up (Federal Government of Germany, 28 July 2022, p. 13; Federal Government of Germany, 6 December 2022, p. 10).

This was a mixed policy. Growth loans after listing widen finance beyond an IPO. But procurement, which it has been argued can create early markets for innovative firms (Edler & Georghiou, 2007), was left to a platform still being prepared, and the listing reform was a set of key points.

3 ANALYSIS OF THE POLICIES

3.1 CONTEXTUAL ANALYSIS: ORIGINS, BACKGROUND AND RATIONALE OF THE POLICIES

Policy in 2022 was made under changes of government. In Korea, where Yoon Suk Yeol's government succeeded Moon Jae-in's during the year, the ministry described its turn from a venture boom grown on government liquidity to private-led investment and global expansion (MSS, 27 December 2022, work plan, p. 2). In France Élisabeth Borne's government was formed in May, with the economy ministry renamed for industrial and digital sovereignty (President of the French Republic, 20 May 2022, Art. 1). The United Kingdom changed its Prime Minister twice: Liz Truss's government presented the Growth Plan on 23 September, and on 25 October Rishi Sunak became Prime Minister and re-appointed Jeremy Hunt, whom Liz Truss had made Chancellor on 14 October (Prime Minister's Office, 14 October 2022, appointments; Prime Minister's Office, 25 October 2022, appointments). China's State Council remained under Premier Li Keqiang (State Council of the People's Republic of China, 1 October 2022, preamble), and Japan's government named 2022 "the first year of start-ups" (Government of Japan, 28 October 2022, p. 34).

The fiscal setting was one of relief giving way to consolidation. China planned a deficit ratio of about 2.8 %, below that of 2021 (State Council of the People's Republic of China, 5 March 2022, Part III (1)), and Japan's economic measures of 28 October put fiscal spending at JPY 39.0 trillion (Government of Japan, 28 October 2022, p. 53). The budget of Korea's ministry for 2023 was passed at KRW 13.5205 trillion, against KRW 18.8412 trillion in its main budget for 2022 (MSS, 26 December 2022, p. 1, and table). The British Autumn Statement reversed nearly all the measures of the Growth Plan (HM Treasury, 17 November 2022, p. 1), and Germany's budget for 2023, passed by the Bundestag on 25 November with net borrowing of EUR 45.610 billion against a maximum of EUR 45.616 billion, met the borrowing limit of the Basic Law for the first time in three years (Bundesrat, 8 December 2022, pp. 25–26). The US appropriations for fiscal year 2023 were enacted on 29 December (U.S. Congress, 29 December 2022, 136 Stat. 4694), and France's Finance Act for 2023 on 30 December (Parliament of France, 30 December 2022, Art. 55).

Each government represented the problem of entrepreneurship in its own way, and the representation shaped the solutions it chose (Bacchi, 2009; Rochefort & Cobb, 1993). Table 2 sets out what each government represented the problem to be, what the representation presupposed and what it left unsaid.

Despite these differences, the governments converged on two rationales: that the shortfall lay in finance, to be closed by funds of funds, guarantees and investor reliefs, and that the cost of founding could be cut by moving formalities online. Entrepreneurship policy has been urged to show such market failures rather than assume them (Acs et al., 2016). Similar instruments also appeared in different countries, as accounts of policy transfer would lead one to expect (Dolowitz & Marsh, 2000): multiple-voting shares were a bill in Korea, key points in Germany and an intention in France; company formation went online in Germany and France; and Japan announced a

guarantee without a personal guarantee in the year in which the British Recovery Loan Scheme let lenders require one.

Table 2: How the governments represented the problem

Country	Problem represented	Presupposition	Left unsaid
Korea	A venture boom grown on government liquidity and the home market (MSS, 27 December 2022)	Private capital and overseas reach will produce global firms	Unicorns doubled while the rank fell; multiple-voting shares still a bill
Japan	Too few start-ups and too little investment, with founders bound by personal guarantees (Council for the Realisation of New Capitalism, 28 November 2022)	A 10-fold rise in investment can be planned	Contracts with young SMEs at about 1 % against 3 %
China	Market entities under pressure, to be relieved so that they keep jobs (State Council of the People's Republic of China, 24 May 2022)	Refunds, guarantees and designated SRDI firms will sustain growth	Equity for founders; the routes through insolvency
United Kingdom	Growth held back by taxes and red tape, then public finances to restore (HM Treasury, 23 September 2022, 17 November 2022)	Tax rates are the binding constraint	Private businesses 1.5 % fewer; research relief for SMEs cut
United States	A boom in new businesses to sustain with capital and contracts (President of the United States, 31 October 2022)	Applications measure entrepreneurship	Small business vendors almost 5.8 % fewer
France	Production taxes and the industrialisation of start-ups (MEFIDS, 26 September 2022, 24 November 2022)	Tax cuts and projects chosen by the state build a sector	Record creations include 641,500 micro-entrepreneurs
Germany	A country that needs more risk-takers and a new founding spirit (Federal Government of Germany, 28 July 2022)	Finance and capital-market law are the main levers	No money of its own; 28 % know of the shorter discharge

Source: IFEED analysis after Rochefort and Cobb (1993) and Bacchi (2009) of the sources cited.

3.2 THEORETICAL ANALYSIS: THEORETICAL VALIDITY AND RELEVANCE

Ecosystem theory sets the first test. It treats productive entrepreneurship as the output of interdependent conditions and holds that the weakest of them limits the system (Ács et al., 2014; Stam, 2015); policy should therefore attend to the connections between actors, not only to transactions with individual firms (Brown & Mason, 2017). The policies of 2022 met this test in places, such as the US lab-embedded programmes and France's university innovation hubs, but failed it in the aggregate: finance dominated, and culture, one of the attributes whose relationships reproduce an ecosystem (Spigel, 2017), received events, awards and proclamations.

On finance, public money that invests alongside private investors is associated with more investment and more successful exits (Brander et al., 2015), while public funds can crowd out private capital (Cumming & MacIntosh, 2006) and public programmes fail when they are too large or too short-lived (Lerner, 2009). Korea's Fund of Funds and TIPS and Germany's Future Fund modules pass this test, and the US Accrual Debenture would if adopted; France's First Factory call,

which chose projects itself, and the 10-month stop on Germany's ZIM do not. A larger exclusion of capital gains on start-up stock raised the funding of eligible firms (Edwards & Todtenhaupt, 2020), which supports Japan's reinvestment relief and the British seed-scheme limits, but these were a proposal and an announcement at the end of the year.

The same tests apply in the other domains (Table 3). The case that a lower cost of failure encourages risky entry (Lee et al., 2007), the evidence that bankruptcy laws friendly to entrepreneurs go with higher entry (Lee et al., 2011) and the evidence that forgiving personal bankruptcy laws go with more self-employment (Armour & Cumming, 2008) support Japan's planned guarantee, France's separation of assets and allowance, China's suspension and deregistration and Korea's simpler restarter evaluation, and weigh against the personal guarantees allowed in the British Recovery Loan Scheme. The evidence on entry regulation (Klapper et al., 2006) supports online formation in Germany and France and weighs against new duties at formation, such as US ownership reporting. The evidence that immigrants are more likely than similarly educated natives to start companies (Hunt, 2011) supports the British routes and weighs against visa reforms left as plans in Korea and Japan.

Table 3: Theoretical validity of the main policies

Theory	Claim	Policies consistent with it	Policies in tension with it
Entrepreneurial ecosystems (Brown & Mason, 2017; Stam, 2015)	The weakest condition binds; relational policy	Lab-embedded programmes (United States); innovation hubs (France)	Global Startup Campus (Japan); SRDI ladder (China)
Government venture capital (Brander et al., 2015; Cumming & MacIntosh, 2006; Lerner, 2009)	Public money works best alongside private investors and when stable	Fund of Funds and TIPS (Korea); Future Fund modules (Germany); Accrual Debenture proposed (United States)	First Factory selection (France); ZIM stop (Germany); Investment Zones (United Kingdom)
Guarantees and failure (Armour & Cumming, 2008; Lee et al., 2007; Lee et al., 2011; Riding et al., 2007)	Guaranteed credit can be incremental; a lower cost of failure goes with more entry	Guarantee planned (Japan); separation of assets (France); deregistration (China); restarter evaluation (Korea)	Personal guarantees allowed (United Kingdom); second chance an aim (Germany)
Investor taxation (Edwards & Todtenhaupt, 2020)	Relief on start-up gains raises funding	Venture and angel conditions (China); reinvestment relief proposed (Japan); seed limits announced (United Kingdom)	Investment tax breaks only proposed (Korea)
Entry regulation (Klapper et al., 2006)	Costly entry deters entry	Online formation (Germany); one-stop shop (France); registration rules (China); start-up scope (Korea)	Ownership reporting (United States); register reform pending (United Kingdom)

Theory	Claim	Policies consistent with it	Policies in tension with it
Talent and education (Hunt, 2011; Kerr & Kerr, 2020; Oosterbeek et al., 2010)	Immigrants start companies out of proportion; education may not raise intentions	Scale-up and High Potential Individual routes (United Kingdom)	Visa plans (Korea, Japan); no founder route (Germany); graduate measures (China)
Demand, payment and markets (Edler & Georghiou, 2007; Petersen & Rajan, 1997; Vismara et al., 2012)	Procurement can create markets; trade credit flows to firms that lack it; second markets for small firms underperform	Trial purchases (Korea); SME preferences and payment terms (China); SBIR extension (United States)	1 % against 3 % (Japan); Procurement Bill pending (United Kingdom)

Source: IFEED assessment of the policies reviewed in Section 2, against the sources named in the first column.

3.3 PRACTICAL ANALYSIS: PERFORMANCE, CONTRIBUTION AND IMPACT

By the end of 2022 much of the year's policy had not yet reached firms (Table 4): the measures in force were concentrated in registration, tax relief and guarantees, and many reforms were plans, tax outlines, key points or bills. Where governments reported performance, they reported outputs: 10,461 Korean restarters who passed the integrity evaluation from 2017 to 2021 (MSS, 15 March 2022, p. 1), more than 95,000 British Start Up Loans since 2012 (BEIS, 25 September 2022, press release), a record USD 154.2 billion of US prime contracts for small businesses (SBA, July 2022, p. 1), nearly 7,400 Bpifrance honour loans in a half-year (Bpifrance, 27 September 2022, p. 11) and 227 German EXIST grants in 2021 (Federal Government of Germany, 6 December 2022, p. 2). The figures that cut against the governments' narratives came from the governments themselves: Korean venture investment in the third quarter was KRW 838.8 billion below a year earlier, after rises in the first two quarters (MSS, 4 November 2022, p. 1); Japanese start-up investment was about JPY 820 billion in 2021, against the plan's aim of more than 10 times that amount (Council for the Realisation of New Capitalism, 28 November 2022, p. 2); British private businesses fell by 1.5 %, or 82,000, to 5.5 million at the start of 2022 (BEIS, 6 October 2022, p. 1); the number of small US federal vendors fell by almost 5.8 % as contract dollars rose (SBA, July 2022, p. 1); state-guaranteed loans through Bpifrance fell from EUR 9 billion to EUR 2.85 billion in the half-year (Bpifrance, 27 September 2022, p. 2); and of 61,000 innovative or growth-oriented German start-ups in 2021, only 12 % sought venture capital (KfW, 28 September 2022, press release).

Contribution and impact were rarely shown. Performance measures serve different purposes — to evaluate, to control, to celebrate or to promote (Behn, 2003) — and those of 2022 served mostly to celebrate and to promote. Yet some designs would allow evaluation: the proposed rule for US development centres would let clients opt in to surveys of outcomes, Japan measured its purchasing from young SMEs against a target, and Germany's Start-up Strategy promised an evaluation of all its measures for start-ups (Federal Government of Germany, 28 July 2022, p. 3). Table 5 summarises the assessments of Section 2 by domain and country.

Table 4: Status of the main measures at the end of 2022

Country	In force or launched in 2022	Adopted, in force from 2023	Pending or announced
Korea	Start-up Support Act; Fund of Funds call; restarter evaluation; KRW 50 million option ceiling; Venture Investment Act decree (due from 23 August)	Ministry budget for 2023 (TIPS, super-gap project); product trials	Multiple-voting shares; Venture Business Act amendment; tax proposals; private funds of funds; visa change; M&A fund limit
Japan	Shoko Chukin Bank's loans without management guarantees; employment-insurance rule for founders (enacted; date of effect not stated)		Security right; guarantee scheme; tax outline; FILP ceiling; JST and NEDO funds; campus; visa; trading rule; procurement target
China	SAMR Order No. 52; Decree No. 755; refunds and re-guarantees; venture and angel conditions; procurement preferences; SRDI measures; graduate notice		Full registration system for share issues
United Kingdom	Finance Act 2022 allowance; Recovery Loan Scheme; Start Up Loans; Help to Grow: Management; Scale-up and High Potential Individual routes		Register reform; SEIS and CSOP limits; R&D relief changes; Investment Zones; Procurement Bill; prospectus reform
United States	Burden-reduction rule; Community Advantage changes; SBIR and STTR extension; CHIPS and Science Act programmes; appropriations for 2023	Payroll offset of the research credit	SBIC, SBLC, affiliation and SBDC proposals; ownership reporting (final rule, effective in a later year)
France	Sole-trader statute; allowance after a non-viable business; one-stop shop opened; value-sharing bonus; First Factory call; French Tech programmes; hubs call	Finance Act for 2023 (levy halved); one-stop shop compulsory	Tech IPO measures; multiple-voting rights
Germany	Online formation; Venture Tech Growth Financing; DeepTech & Climate Fund; INVEST; EXIST; ZIM reopened	Budget for 2023 (passed by the Bundestag); formation with contributions in kind	Future Financing Act; sandbox act; skilled-immigration reform; EXIST Women; procurement platform

Source: IFEED compilation from the sources cited in Section 2. An empty cell: no measure of that status in Section 2. A final rule that takes effect after 2023 is shown, with its status, in the last column.

Table 5: Assessment of the policies by domain and country

Domain	Korea	Japan	China	United Kingdom	United States	France	Germany
Regulation and Law	Mixed	Mixed	Good	Poor	Mixed	Good	Mixed
Finance	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed
Culture	Mixed	Mixed	Poor	Poor	Poor	Mixed	Mixed
Support	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed
Human Capital	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed
Market	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed	Mixed

Source: IFEED assessment of the policies reviewed in Section 2, on the sources cited there. Where a subsection assessed its policies differently, Mixed is shown.

4 POLICY LIMITATIONS

4.1 POLICY DEBATES, CONTESTED POINTS AND ISSUES

The policies of 2022 raised six points of contention, set out in Table 6. The first two concerned capital and control: whether public funds will complement or displace private investors, which turns on who selects the firms, and whether shares with extra votes protect founders at the expense of outside shareholders. Venture capital has been found to spur patenting far more than corporate research (Kortum & Lerner, 2000), which is why the design of public funds matters.

The third and fourth concerned failure and people: whether second chances weaken the protection of creditors, and whether migration control should override the supply of founders and staff. The fifth and sixth concerned rules and money: whether relief from formalities can coexist with new duties, and whether incentives that work only if they are stable are compatible with reversed plans, closed programmes and temporary reliefs.

Table 6: Debates and contested points in the policies of 2022

Issue	What the governments set out	Point at issue
Public capital	Fund of Funds (Korea); growth loans with private lenders (Germany); First Factory call (France); SRDI ladder (China) (KfW, 1 August 2022; MIIT, 1 June 2022; MSS, 22 February 2022; MEFIDS, 24 November 2022)	Complement or displacement; who selects
Founders' control	Multiple-voting shares as a bill (Korea), key points (Germany) and an intention (France) (Federal Ministry of Justice, 29 June 2022; MSS, 12 July 2022; MEFIDS, 23 November 2022)	Founder control against outside shareholders
Cost of failure	Guarantee without a personal guarantee (Japan); separation of assets (France); personal guarantees allowed (United Kingdom) (BEIS, 20 July 2022; METI, FSA, & Ministry of Finance of Japan, 23 December 2022; Parliament of France, 14 February 2022)	Second chances against creditor protection
Talent	Scale-up route (United Kingdom); visa plans (Korea, Japan); immigration key points (Germany) (Council for the Realisation of New Capitalism, 28 November 2022; Federal Government of Germany, 30 November 2022; Home Office, 15 March 2022; MSS, 8 September 2022)	Migration control against the supply of talent
Burden relief and new duties	Online formation (Germany); one-stop shop (France); ownership reporting (United States); identity checks (United Kingdom) (BEIS, 28 February 2022; Federal Ministry of Justice, 29 July 2022; FinCEN, 30 September 2022; MEFIDS, 16 November 2022)	Relief against new obligations
Stable incentives	Growth Plan reversed (United Kingdom); ZIM stop (Germany); temporary set-aside (China); investment tax breaks proposed (Korea) (Federal Government of Germany, 6 December 2022; HM Treasury, 17 November 2022; MOEF, 21 July 2022; MOF, 30 May 2022)	Revenue against stable incentives

Source: IFEED analysis of the sources cited.

4.2 THEORETICAL LIMITATIONS OF THE POLICIES

The first limitation is that the policies treated capital as the binding condition everywhere. Ecosystem theory holds that the weakest condition limits the system (Ács et al., 2014), but the governments assumed more than they showed. Their own figures point in more than one direction: only 12 % of innovative or growth-oriented German start-ups sought venture capital (KfW, 28 September 2022, press release), only 25 % of people in Japan see starting a business as a desirable career (Council for the Realisation of New Capitalism, 28 November 2022, p. 4), and about half of Japanese founders who borrowed from private banks had given a personal guarantee (FSA, 2 November 2022, p. 26).

The second is the selection of firms by the state. China's SRDI ladder, Korea's super-gap project, France's Next40/120 and First Factory call choose firms or projects by status, field or past fundraising, but it has been argued that high-growth firms are too varied to be identified in advance (Mason & Brown, 2013), and large public programmes have failed when they could not let markets choose or stop what did not work (Lerner, 2009).

The third is an asymmetry in the treatment of failure. Japan, France, China and Korea acted on guarantees, personal assets, exit and restarts, although Japan's guarantee was announced to start in March 2023; the United Kingdom let lenders in its main guarantee scheme require personal guarantees, and Germany's second chance remained an aim while only 28 % of the working population knew of the shorter discharge (KfW, May 2022, p. 11). Bankruptcy laws that are friendly to entrepreneurs go with higher entry (Lee et al., 2011), but it has been argued that stigma can blunt their effect (Lee et al., 2007), and stigma was addressed by events and statements.

The fourth is the weakness of evaluation. Entrepreneurship policy has long been criticised for subsidising the number of start-ups rather than their quality and for lacking evaluation (Acs et al., 2016; Shane, 2009). The figures of 2022 — applications counted as businesses, loans made, funds offered, incubators recognised — repeat the counting, and few of the designs came with a plan to measure their effects.

5 POLICY IMPLICATIONS

First, public capital should follow private choices. The designs that the evidence supports — funds of funds that commit to private managers, grants that follow private investment, loans made alongside private lenders — were present in 2022 in Korea and Germany and proposed in the United States, and should be the rule rather than the exception. The United States should finalise its rule for investment companies, Korea should keep TIPS at the scale of its adopted budget and the Fund of Funds behind private managers, and France and China should leave more of the selection of firms to private investors.

Second, reforms should be brought into force. Korea should enact multiple-voting shares; Japan should submit its bill on the business-growth security right and enact its tax outline; the United Kingdom should legislate its register reform and pass the Procurement Bill; Germany should turn the key points of its Future Financing Act into law; and France should begin the legislative work on multiple-voting rights that it announced.

Third, the cost of failure should be lowered where it bites: personal guarantees, the personal assets of sole traders and the records that follow a failed founder. Japan should launch its guarantee on time and report whether the loans are incremental, the United Kingdom should reconsider personal guarantees in a scheme backed by public money, Germany should make its shorter discharge known, and Korea and China should extend their restart and exit rules into measures on stigma.

Fourth, the doors for foreign founders and the staff of growing firms should be kept open. Korea should revise its technology start-up visa as it announced, Japan should let private bodies confirm start-up visas, Germany should add a route for founders to its immigration reform, and the United Kingdom should publish the immigration plan for growth that it promised.

Finally, policy should be stable and evaluated. Reversed plans, programmes closed to applications and temporary reliefs cannot change decisions that run over years. Governments should report purchasing from young firms every year, as Japan began to do, and should commit, when they adopt measures that select firms by rules, to evaluating them against what would have happened without them and to publishing the results, as Germany's Start-up Strategy promised.

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